

Yousuf Eslam

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EDUCATION

Ain Shams University

Mechatronics and Robotics Engineering

Oct. 2023 – Jul.

Cairo, Egypt

- **CGPA:** 3.02(Very Good)

EXPERIENCE

YouTube

Dec. 2024 – Present

Creator (@YousufEslam)

- Grew channel to **8K Views, over 600 watch time(hours) in 1.5 years**; created 30+ on STEM subjects and Project Documentaries.
- Explained engineering topics such as: Fluid Mechanics, Thermodynamics, Material Science, Heat Transfer, etc
- **Highlights:** [Air Powered Car Project Vlog](#), [How I Make Videos](#), [Intro to Vehicle Dynamics](#)

Aerodynamics Engineer

Nov 2024 – Present

ASU Racing Team, Formula Student

- **Designed and validated the vehicle bodywork** using Fusion 360 and Ansys Fluent according to biomimcry principles(referenced to a white shark and a swift bird.)
- **Designed and validated aerodynamic wings** for better traction.
- Designed Sheet metal fixations and checked with stress analysis.
- Tutored "Intro to Aerodynamics" and "Intro to VHD" in the recruitment academy.

Renewable Energy Intern

Jul. 2025 – Aug. 2025

NREA

Cairo, Egypt

- Intern at the New and Renewable Energy Authority(NREA), got to see different solar and wind energy equipment at the Authority's HQ.

Nuclear Energy Intern

Aug. 2025

EAEA

El Sharqia, Egypt

- Intern at the Egyptian Atomic Energy Authority(EAEA), **visited the second Egyptian power plant.**
- Learned mechatronics implementations of nuclear energy. ex: Radiation Sensor Devices, PCB Fabrication, Nuclear Plant Control Room, etc..

Data Analytics and Visualization Course Attendee

Nov. 2023

University of Illinois Urbana-Champaign(Online)

Cairo, Egypt

- Applied based data analysis principles.
- Visualized data using Tableau.

PROJECTS

Air Powered Car

- * Led a team of 8 in designing a lightweight, functional compressed-air car powered by a thrust nozzle. The system operated at 10 bars with a 3-liter air reservoir and was designed to travel 10 meters. The vehicle also incorporated braking and remote-control capabilities.

Hydraulic Clutch System

- * applied fundamental fluid mechanics principles to design and build a functional fluid clutch system from scratch. A basic DC motor drove a primary impeller, which transmitted rotational motion through a liquid medium (water, in our case) to a secondary impeller connected to a pulley.

RC Car

- * Implemented features such as hand-gesture control, Bluetooth button operation, and line-tracking capability.

Omni-Wheel Mobile Manipulator

- * Designing an autonomous mobile robot arm, integrating Machine Learning, Model Based Design and Ros 2.

SKILLS

Engineering Tools : Inventor, Fusion 360, SolidWorks, SpaceClaim, MATLAB, Ansys(Stress analysis and CFD), Tableau(Data analysis), Obsidian

Programming Languages : Java, C++, Python , Arduino

Graphics Design Tools : Blender 3D, Davinci Resolve, OBS, Photoshop

Languages: Arabic(Native), English(Fluent), French(Beginner).